Sean Potts, r/wallstreetbets API calling Final Project

Research Question: How has a stock's performance influenced their discussion on reddit over time?

Who Cares: Almost everyone cares about their finances and the economy, now more than ever. Specifically, reddit users have been known to influence stock trading before and there is a large community of people on reddit who come there just to discuss finances and stock options like on r/wallstreetbets or other subreddits. This is the same subreddit who influenced the GameStop short back in early 2021. There is a large community of online investors and financial institutions who would find this information useful as a metric to determine what stocks are more popular than others.

What I expect to find: I'm hoping I'll find data on each stock over time as well as comment data to show which ones have been talked about more frequently recently and which ones have done well in the past. I'm hoping this will give me some insight into what stocks tend to perform well and which ones people are keeping their eye on.

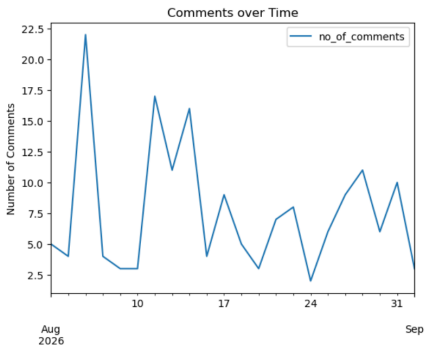
The exact Data Source (with links): The data source comes from the Tradestie API calling from reddit. The exact link is: https://tradestie.com/api/v1/apps/reddit. This is the url that I called from, though I added an optional API key in the parameters to be able to get more requests per minute.

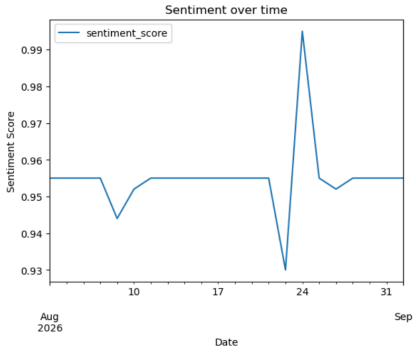
How I will get it: I'll call the data from the api and then format it as json. From there, I'll have to specify the dates and stocks that I want to ask about as well as their features such as if they are "bullish" or "bearish" and how many comments each one has. After that, I can convert it into a DataFrame with pandas and plot that data with Matplotlib.

The scope(what records, over what time, etc…): I want to get about 10 years worth of data so I was hoping I'd be able to call all the entries from 2016-now.

The analysis: charts, comparisons, data that answers the question

These show the number 1 performing stock over time by sentiment score. The line isn't one particular stock, but just the best stock that day. They are top performing by sentiment score and we can compare that to the number of comments that it gets in the graph on the left.





Terms of service and feasibility: Since my data is coming from a free, publicly available API, it is entirely ethical to pull from. Additionally, none of the entries that I am taking have personally identifiable information such as user data or any other personal data. The API has very little documentation on ethics or terms of service, so I am assuming that anything I query for qualifies for fair use. To make sure I am staying ethical in my data collection, I do specify my user agent in the headers with my CU email and the name of this class so the API knows who is calling on it.

One record: See photo on left

←—

```
print(response.status_code)
```

```
200
```

```
resp = response.json()
```

```
len(resp)
```

```
50
```

```
resp
```

```
[{'ticker': 'AI',
  'sentiment': 'Bullish',
  'sentiment_score': 0.114,
  'no_of_comments': 101},
 {'ticker': 'SPY',
  'sentiment': 'Bullish',
  'sentiment_score': 0.5,
  'no_of_comments': 50},
 {'ticker': 'MU',
  'sentiment': 'Bullish',
  'sentiment_score': 0.226,
  'no_of_comments': 45},
 {'ticker': 'AMD',
  'sentiment': 'Bullish',
  'sentiment_score': 0.057,
  'no_of_comments': 39},
 {'ticker': 'META',
  'sentiment': 'Bullish',
```

Risks and Solutions: The biggest risk I have to finish is perhaps the scale of my data. I might end up finding that there aren't any particular stocks that stand out or maybe that the length of time that I've picked doesn't show any clear trends. If I run into this problem though, I'll probably just end up narrowing my scope to a window of time that is more recent or even cutting my window in half.

AI Disclaimer: Most of my code is my own, however the API was very particular about how I could only call one date per request, so I had to make a loop for dates in a specified range. I made the range bit myself, but I had Claude make me a loop that also told me that I was getting 400 errors from the api whenever I ran it. To solve this, I just asked the API's support hotline what to do and they said it was probably the date format that was causing the issue. I also had Claude make a test graph that I didn't end up using.